

CM515_Assignment_13

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Instructions

The aim of this assignment is to provide you with an opportunity to sharpen your skills in using `ggplot()`. While some of the plots you create may resemble those we covered in class, it is essential that they are original.

Rules for the Assignment:

1. All data sets except iris are fair game. That includes past data sets used in the class, sets built into R, your own data, and even online data.
2. All graphs must include axis labels, plot title, a theme of your choice, and a brief description/interpretation of the plot (2-3 sentences).

It's worth noting that `ggplot()` is a widely-used tool, and there are numerous online resources available for you to explore. We strongly recommend that you take advantage of these resources to deepen your understanding of `ggplot()`.

Knit the document into a PDF and submit it to Canvas by 04/27/2026 at 11:59 pm.

Load Packages and Data

1. Make a Scatter Plot With Customized Point Size and Transparency (3pts)

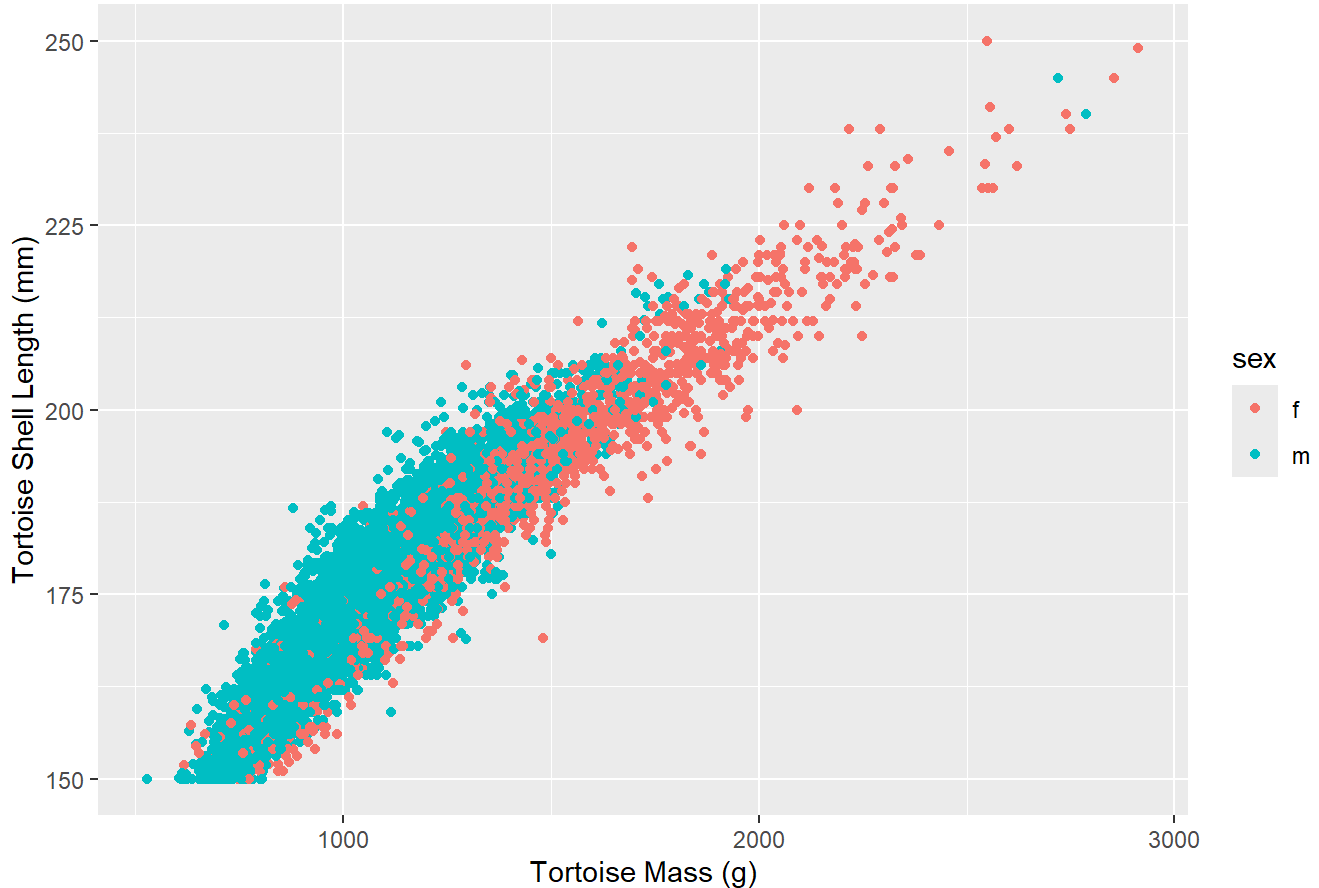
Description and Interpretation:

In many species the males tend to be larger than their female counter parts. I wanted to see if that was the case in this study. According to this graph the females of this species appear to be larger in mass and shell length than their male counter parts.

individual <chr>	y... <int>	year_recode <int>	sea... <chr>	locality <chr>	s.. <chr>	body_mass_gra... <dbl>	body_condition_index <dbl>
1 1	2008	1	Summer	Plateau	m	1230	6.717641
2 1	2009	2	Summer	Plateau	m	1271	6.901607
3 1	2010	3	Spring	Plateau	m	1239	6.672052
4 1	2011	4	Spring	Plateau	m	1205	6.457663
5 1	2011	4	Summer	Plateau	m	1300	7.131103

individual <chr>	y... <int>	year_recode <int>	sea... <chr>	locality <chr>	s.. <chr>	body_mass_gra... <dbl>	body_condition_index <dbl>
61	2014	7	Summer	Plateau	m	1269	6.840970

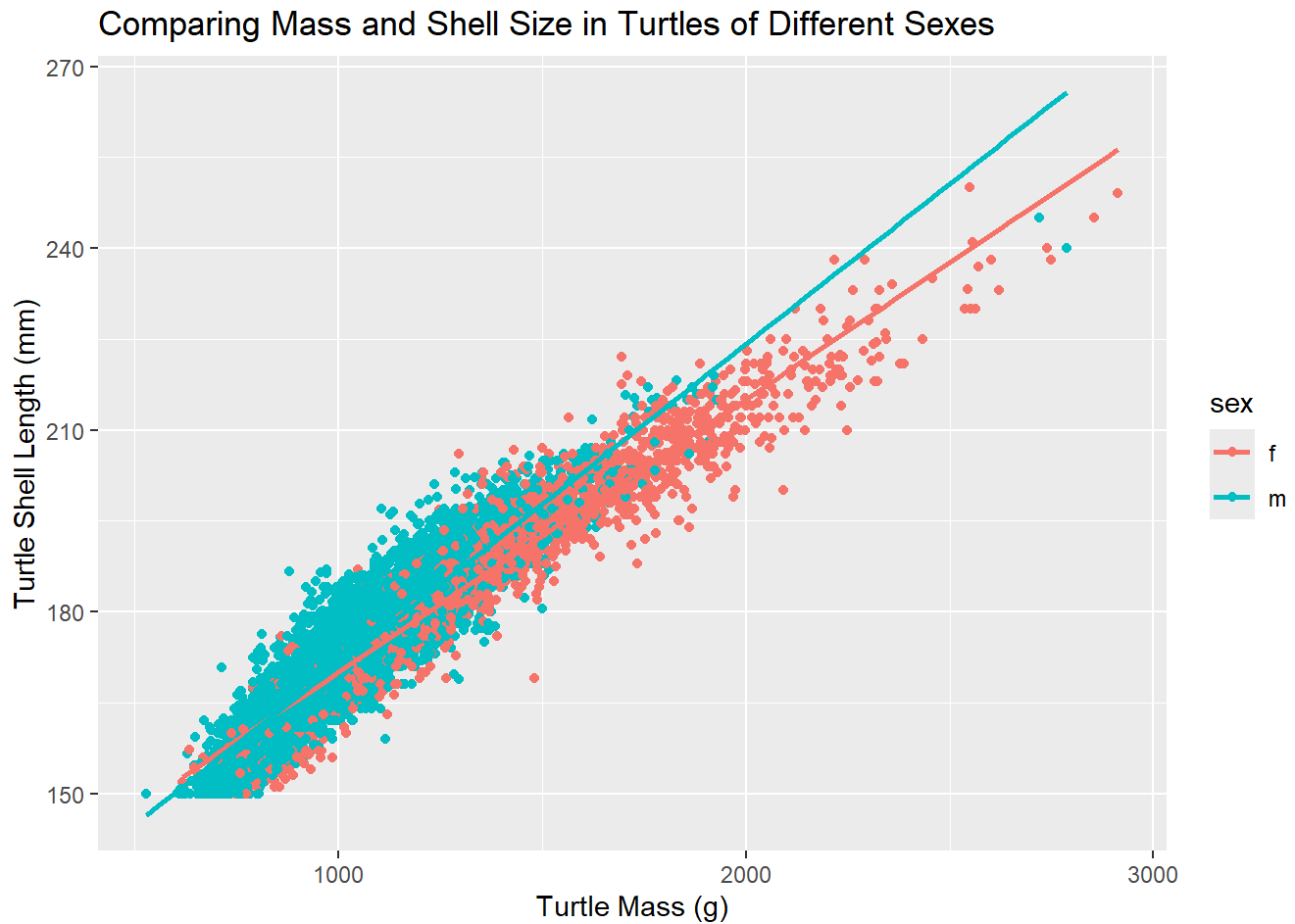
Comparing the Mass and Shell size in Tortoises of Different Sex



2. Fit a Line Through Your Scatter Plot From 1. (3pts)

Description and Interpretation:

Both fit lines (male and female) show a positive relationship between body mass and shell length. As their body mass increases so does their shell length, which is to be expected. However, the male fit line is above the female fit line, which means that for any given body mass, the males have a longer shell length than the females. This indicates that males are more streamlined or have a different shape, while females of the same mass might be denser or have a more voluminous/domed shell.

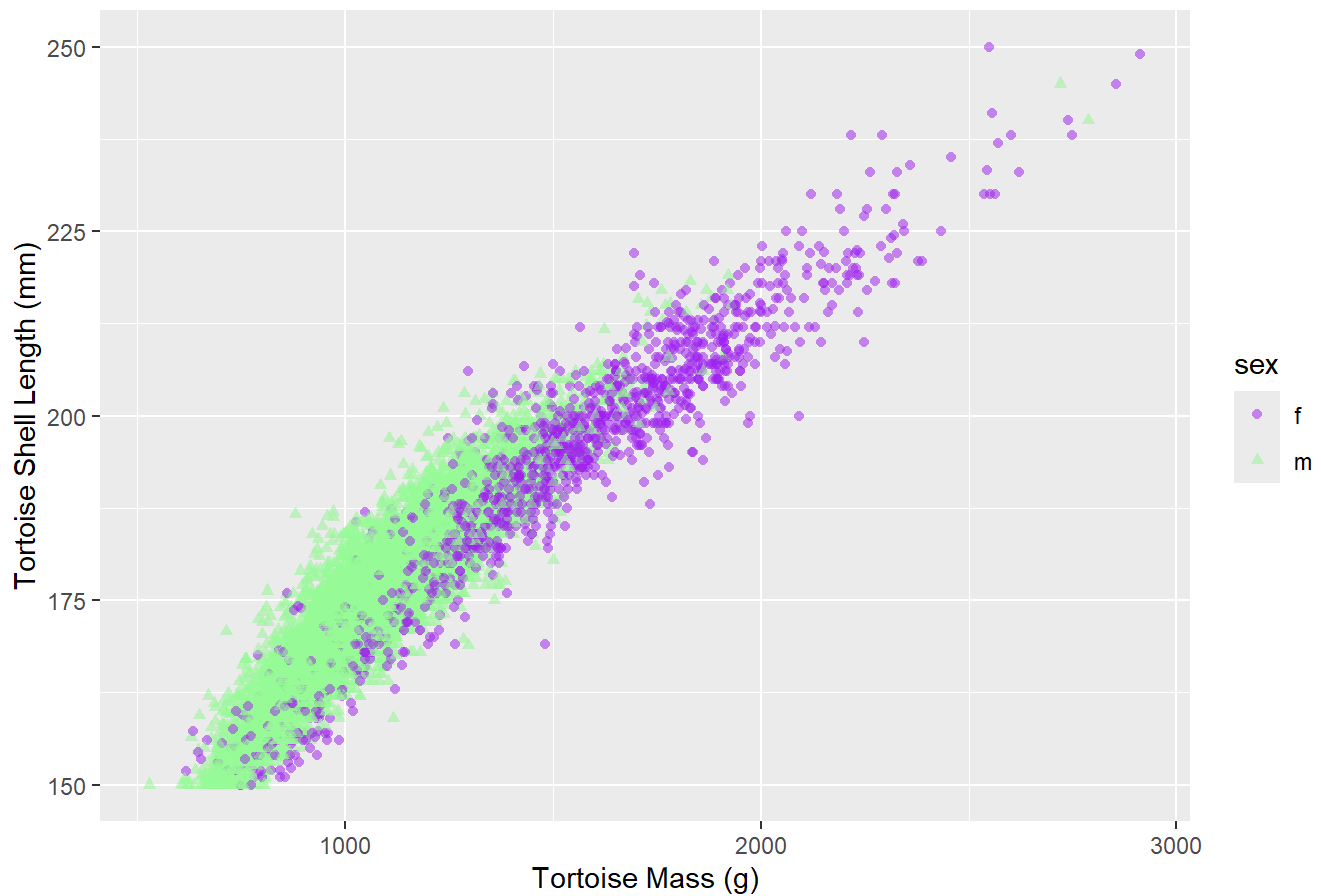


3. Set the transparency, shape, and color of your graph from 1 based on expressions. Scale the color. (3pts)

Description and Interpretation:

I chose an alpha of 0.5 so that the points were not over-shadowing one another too much. I also changed the colors to purple and pale-green so that I was easier to distinguish between values of male tortoises vs female tortoises.

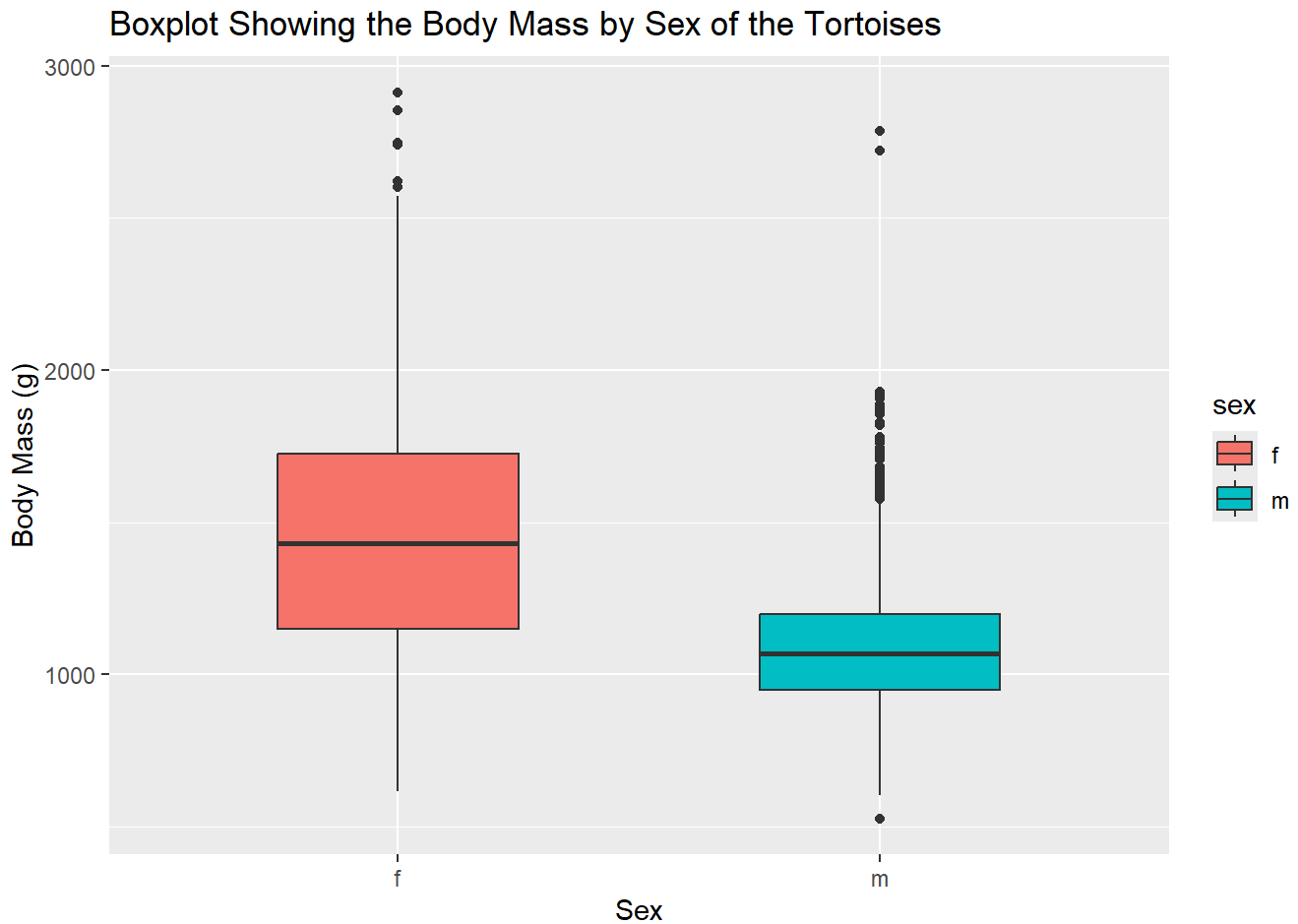
Mass and Shell Size in Tortoises



4. Make a Boxplot With Customized Boxplot Width (3pts)

Description and Interpretation:

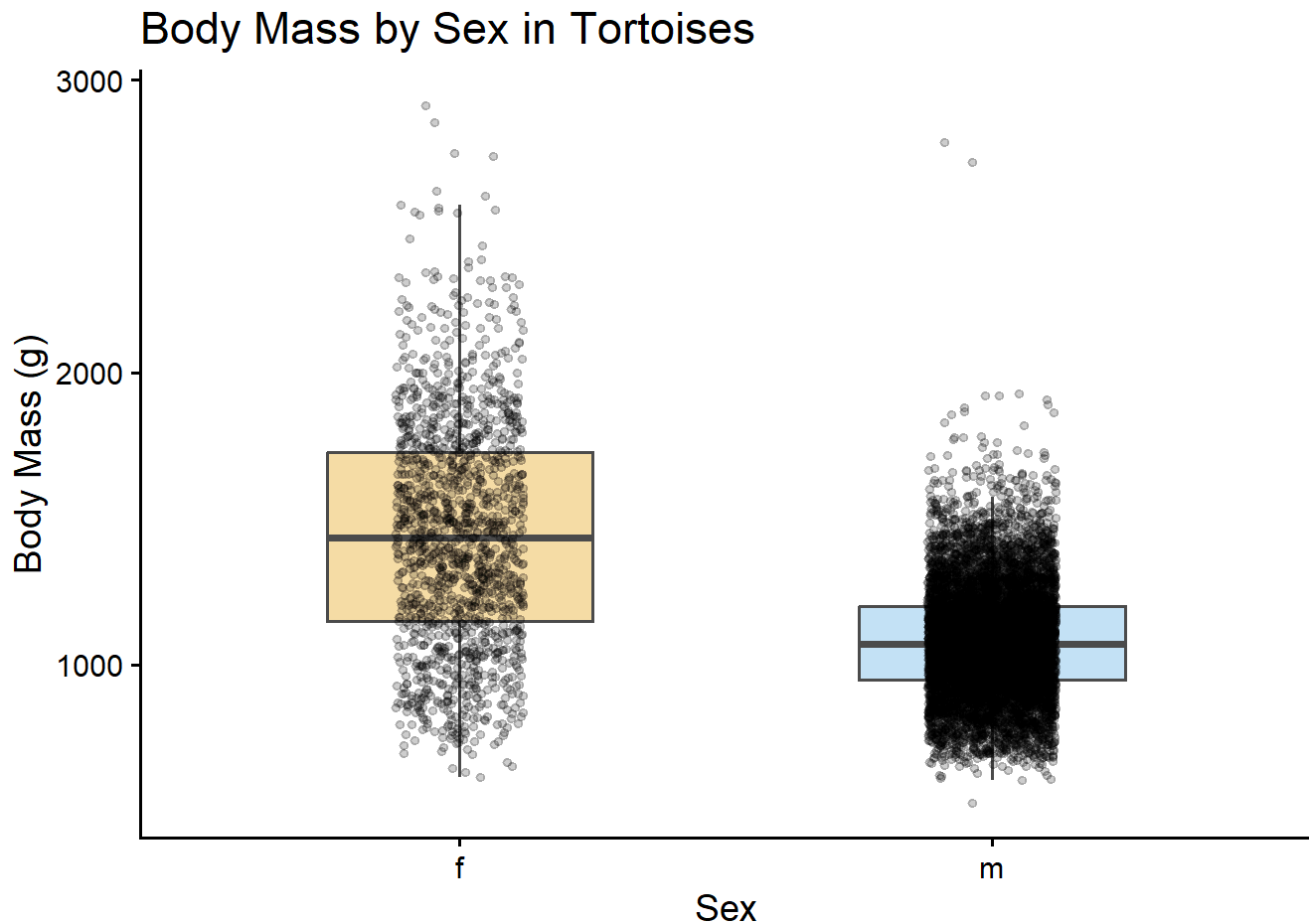
The boxplot provides a concise summary of the body mass distribution for both male and female tortoises, highlighting key differences in their variability. A higher median in the female tortoises suggests sexual dimorphism in size, while the differences in box indicates that one sex (females) has a more consistent growth pattern than the other (male).



5. Overlay the Individual Points Over Your Box Plot From 3. and Adjust the Point Size and Transparency as Needed (3pts)

Description and Interpretation:

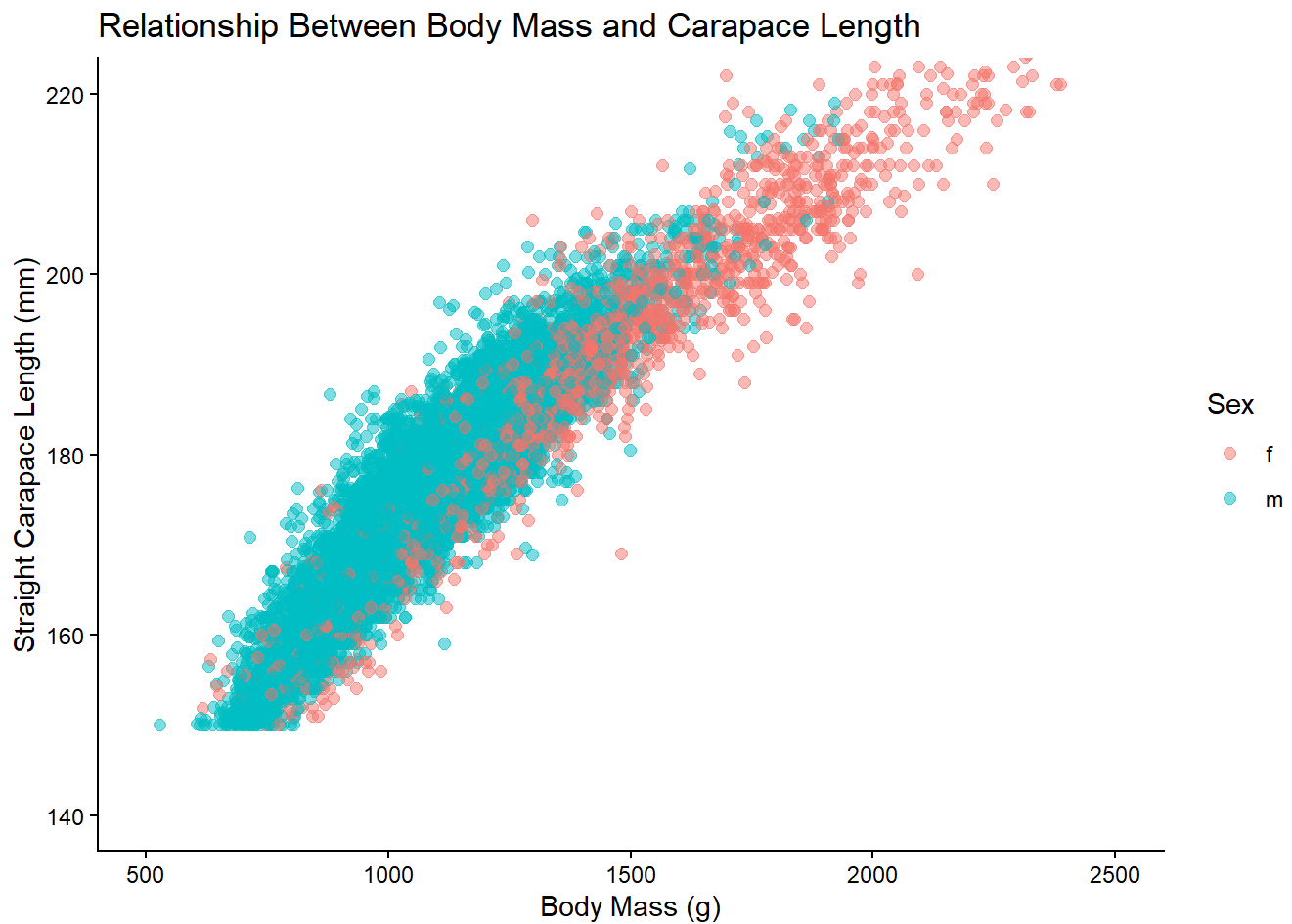
This boxplot with jittered points compares body mass between male and female tortoises. The female group appears to have a higher median body mass and a wider spread, suggesting that females are generally larger and more variable in weight than males.



6. Make a plot and modify the scale in a beneficial way using `scale_x_continuous()`, `scale_y_continuous()`, and `coordinate_cartesian()`. Explain how your modifications to the scales improved the plot. (4pts)

Description and Interpretation:

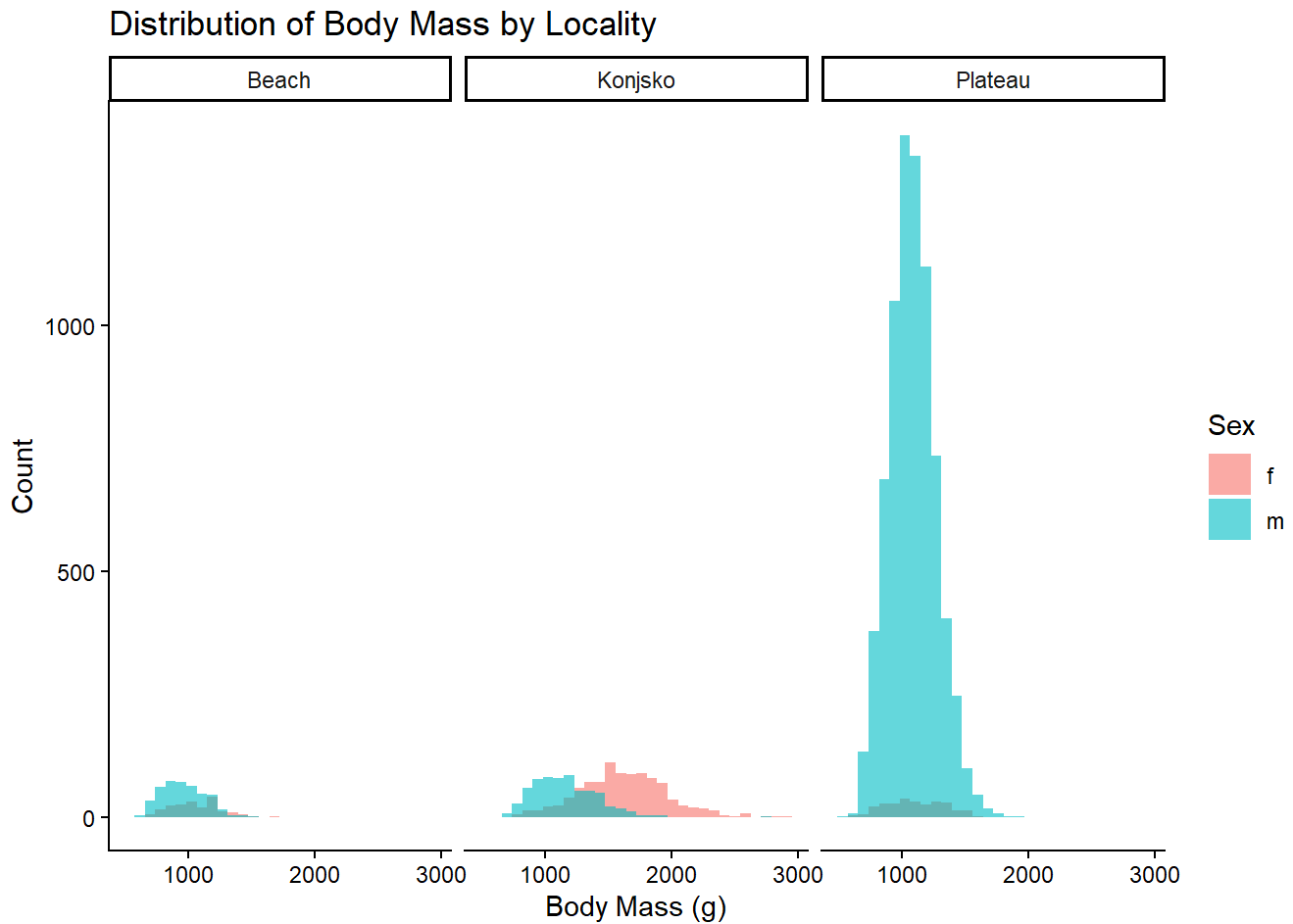
The plot shows a positive relationship between body mass and straight carapace length, meaning larger tortoises tend to have longer shells. Zooming in on the central range makes that trend easier to see because a few very large individuals no longer stretch the axes and compress the rest of the points.



7. Make a New (don't use any plots from 1-4) Plot with `facet_wrap()` (4pts)

Description and Interpretation:

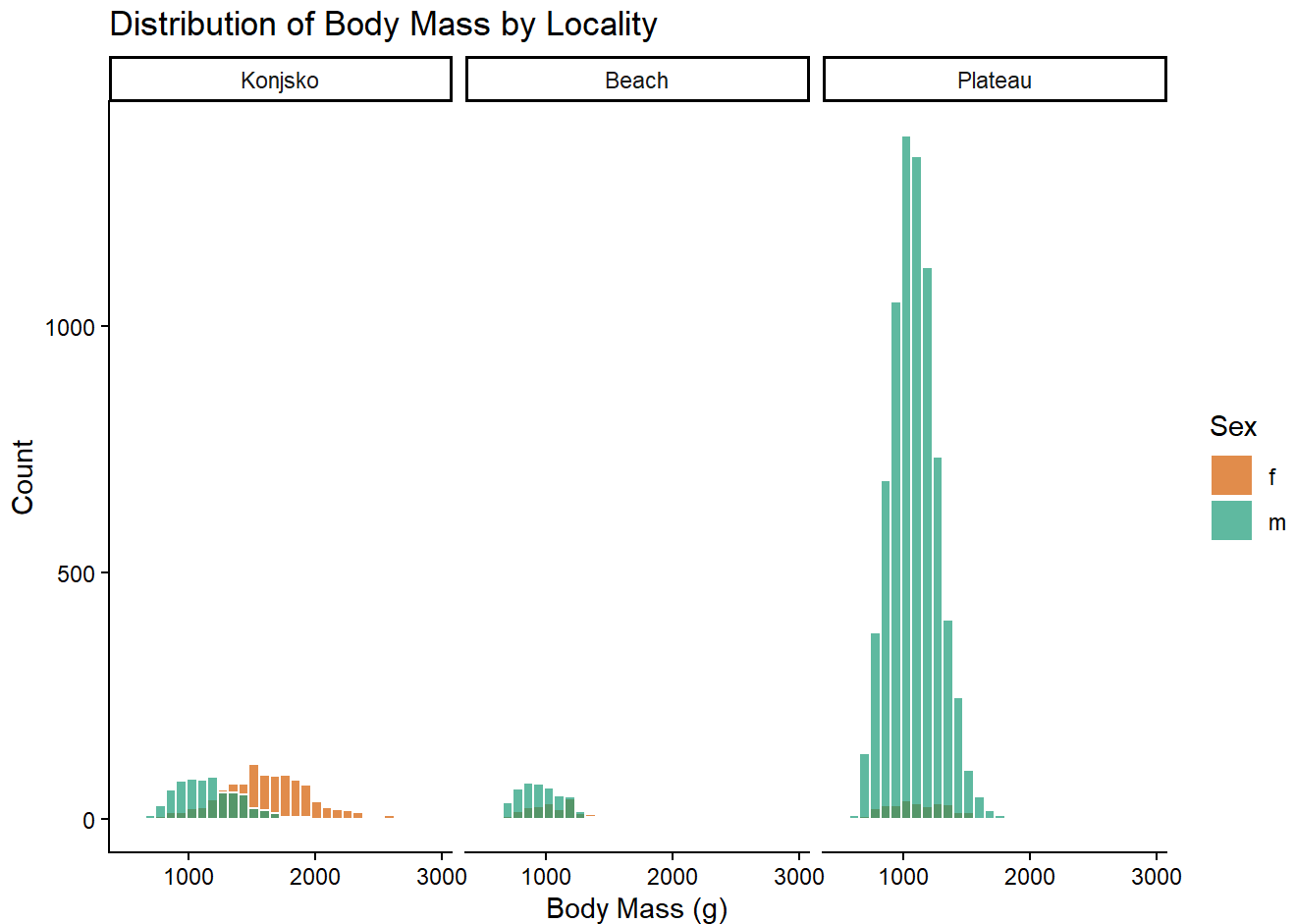
This plot shows the distribution of tortoise body mass within each locality by sex, with separate panels making it easy to compare patterns between groups. It suggests that the Plateau tends to have more tortoises with a greater body mass than the other two locations.



8. Using the plot from 7, scale the colors and reorder your facet to reverse order. (4pts)

Description and Interpretation:

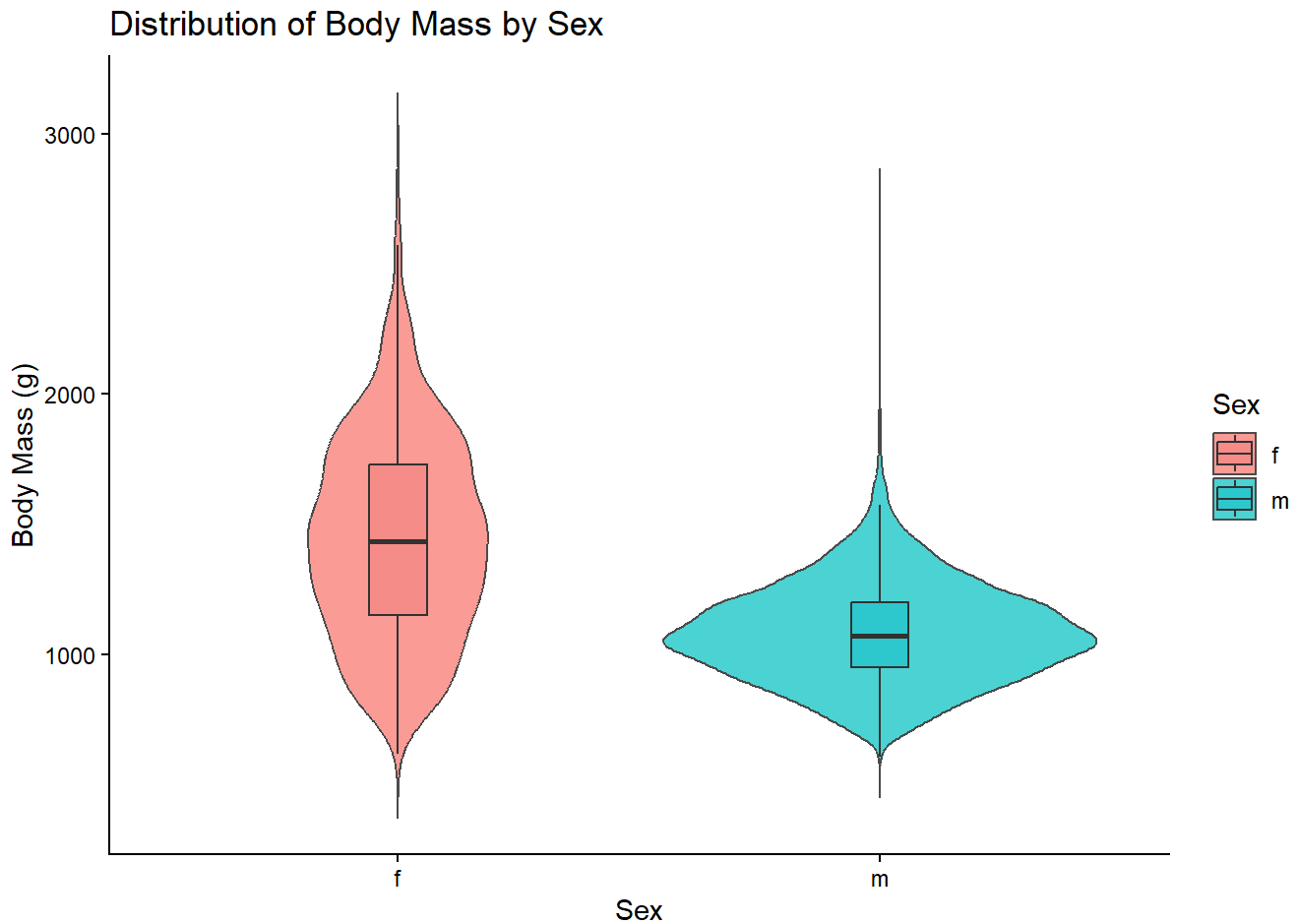
Reversing the facet order makes the panels appear in the opposite direction, and the color scale helps distinguish males from females more clearly. The plot makes it easier to compare whether one locality has generally larger or smaller tortoises and whether the sex distribution differs between locations.



9. Make a Plot Using a Geometry That Was Not Covered in Class (4pts)

Description and Interpretation:

This violin plot compares the distribution of body mass between female and male tortoises. The width of each violin shows where the data are most concentrated, based on the plot the body mass of male tortoises are concentrated around 1000 g where as the body mass of the female tortoises are more varied/spread out. This plot suggests that females tend to have higher body mass overall than males, though there is still overlap between the two groups.



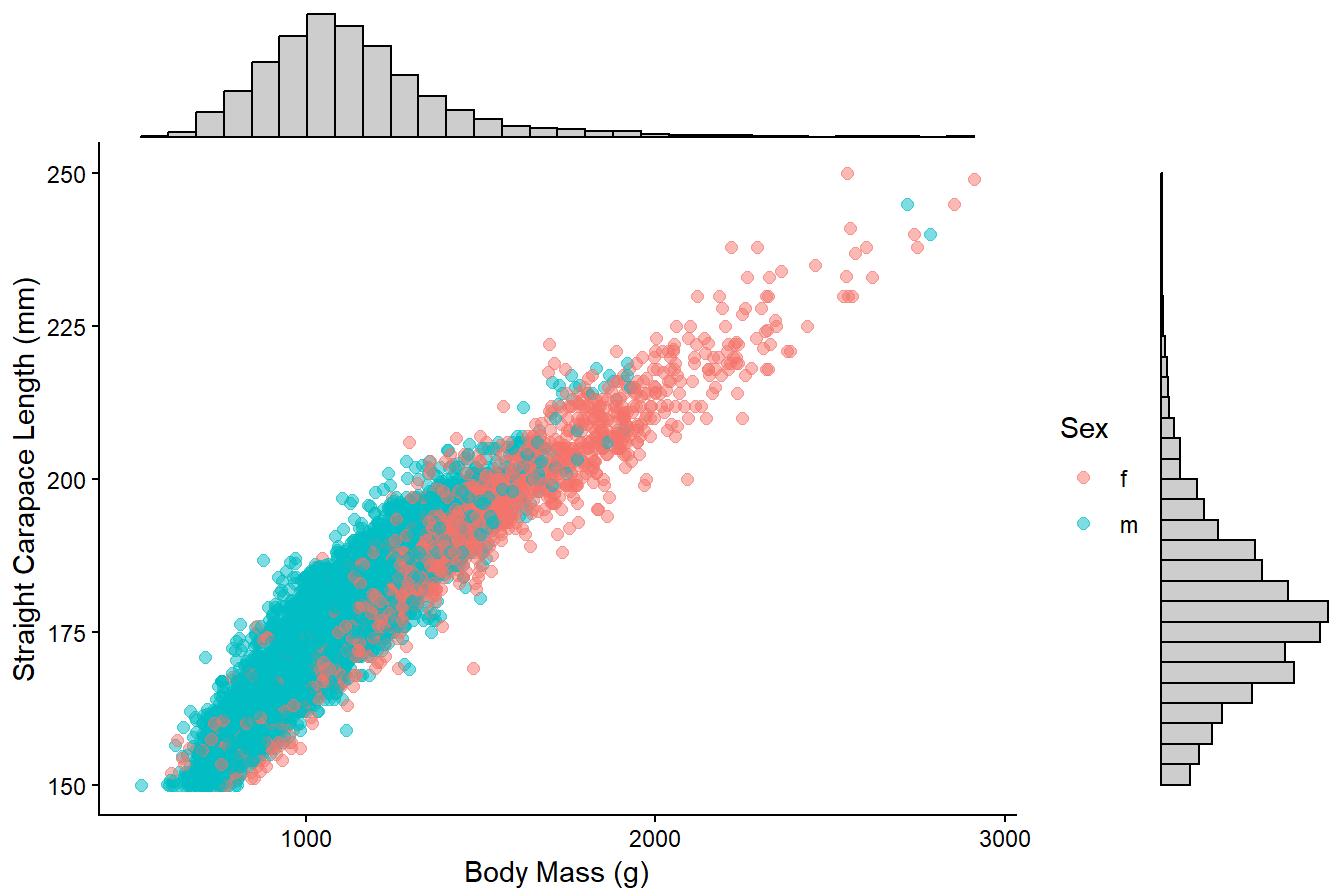
10. Make a Plot Using a ggplot Extension That Was Not Covered in Class. Explain what the extension does. (4pts)

Description and Interpretation:

`ggMarginal()` is a `ggplot2` extension that adds marginal plots around a scatterplot.

This plot shows the relationship between body mass and straight carapace length in tortoises, separated by sex. The marginal histograms show how the two variables are distributed overall, body mass on the top and shell length to the right, which makes the figure more informative than a scatter plot alone. It suggests that larger tortoises tend to have longer shells.

Body Mass and Carapace Length in Tortoises



Appendix

```

knitr::opts_chunk$set(echo = F, message = F, warning = F)

####
#00#
####
library(knitr)
library(tidyverse)
library(ggplot2)

####
#01#
####

clean_tortoise_data <- read.csv("cleaned_tortoiseBC.csv")

head(clean_tortoise_data)

tortoise_data <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams, y = straight_carapace_length_mm, color= sex)) +
  geom_point()+
  labs(title = "Comparing the Mass and Shell size in Tortoises of Different Sex",
        x = "Tortoise Mass (g)",
        y = "Tortoise Shell Length (mm)")

tortoise_data

####
#02#
####
#your code here

?geom_smooth()

tortoise_data_2 <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams,
              y = straight_carapace_length_mm,
              color = sex)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Comparing Mass and Shell Size in Turtles of Different Sexes",
    x = "Turtle Mass (g)",
    y = "Turtle Shell Length (mm)"
  )

tortoise_data_2

####
#03#
####

```

#your code here

```
tortoise_data_3 <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams,
             y = straight_carapace_length_mm,
             color = sex,
             shape = sex)) +
  geom_point(alpha = 0.5) +
  scale_color_manual(values = c("f" = "purple", "m" = "palegreen")) +
  labs( title = "Mass and Shell Size in Tortoises",
        x = "Tortoise Mass (g)",
        y = "Tortoise Shell Length (mm)")
```

tortoise_data_3

####

#04#

####

#your code here

```
tortoise_boxplot <- clean_tortoise_data %>%
  ggplot(aes(x = sex, y = body_mass_grams, fill = sex)) +
  geom_boxplot(width = 0.5) +
  labs(
    title = "Boxplot Showing the Body Mass by Sex of the Tortoises",
    x = "Sex",
    y = "Body Mass (g)"
  )
```

tortoise_boxplot

####

#05#

####

#your code here

```
tortoise_overlay <- clean_tortoise_data %>%
  ggplot(aes(x = sex, y = body_mass_grams, fill = sex)) +
  geom_boxplot(width = 0.5, alpha = 0.35, outlier.shape = NA, color = "gray30") +
  geom_jitter(width = 0.12, height = 0, size = 1.2, alpha = 0.18, color = "black") +
  scale_fill_manual(values = c("f" = "#E69F00", "m" = "#56B4E9")) +
  labs(
    title = "Body Mass by Sex in Tortoises",
    x = "Sex",
    y = "Body Mass (g)"
  ) +
  theme_classic(base_size = 14) +
  theme(legend.position = "none")
```

tortoise_overlay

####

#06#

####

#your code here

```
tortoise_scale_plot <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams, y = straight_carapace_length_mm, color = sex)) +
  geom_point(alpha = 0.5, size = 2) +
  labs(
    title = "Relationship Between Body Mass and Carapace Length",
    x = "Body Mass (g)",
    y = "Straight Carapace Length (mm)",
    color = "Sex"
  ) +
  scale_x_continuous(breaks = seq(500, 2500, by = 500)) +
  scale_y_continuous(breaks = seq(140, 220, by = 20)) +
  coord_cartesian(xlim = c(500, 2500), ylim = c(140, 220)) +
  theme_classic()
```

tortoise_scale_plot

#07#

####

#your code here

```
tortoise_facet <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams, fill = sex)) +
  geom_histogram(bins = 30, alpha = 0.6, position = "identity") +
  facet_wrap(~ locality) +
  labs(
    title = "Distribution of Body Mass by Locality",
    x = "Body Mass (g)",
    y = "Count",
    fill = "Sex"
  ) +
  theme_classic()
```

tortoise_facet

#08#

####

#your code here

```
tortoise_reorder <- clean_tortoise_data %>%
  mutate(locality = factor(locality, levels = rev(unique(locality)))) %>%
  ggplot(aes(x = body_mass_grams, fill = sex)) +
  geom_histogram(bins = 30, alpha = 0.7, position = "identity", color = "white") +
  facet_wrap(~ locality) +
  scale_fill_manual(values = c("f" = "#D95F02", "m" = "#1B9E77")) +
  labs(
    title = "Distribution of Body Mass by Locality",
    x = "Body Mass (g)",
    y = "Count",
```

```

    fill = "Sex"
  ) +
  theme_classic()

tortoise_reorder

####
#09#
####
#your code here

?geom_violin

tortoise_violin <- clean_tortoise_data %>%
  ggplot(aes(x = sex, y = body_mass_grams, fill = sex)) +
  geom_violin(trim = FALSE, alpha = 0.7, color = "gray30") +
  geom_boxplot(width = 0.12, outlier.shape = NA, alpha = 0.4) +
  labs(
    title = "Distribution of Body Mass by Sex",
    x = "Sex",
    y = "Body Mass (g)",
    fill = "Sex"
  ) +
  theme_classic()

tortoise_violin

####
#10#
####
#your code here

library(ggplot2)
#install.packages("ggExtra")
library(ggExtra)

New_extension <- clean_tortoise_data %>%
  ggplot(aes(x = body_mass_grams, y = straight_carapace_length_mm, color = sex)) +
  geom_point(alpha = 0.5, size = 2) +
  labs(
    title = "Body Mass and Carapace Length in Tortoises",
    x = "Body Mass (g)",
    y = "Straight Carapace Length (mm)",
    color = "Sex"
  ) +
  theme_classic()

ggMarginal(New_extension, type = "histogram", fill = "gray70", alpha = 0.6)

```